

Run Accounting and Mission Outcome

Complete campaign: 360 valid, 360 successful, zero exclusions

Success is constant in the current campaign. This supplement documents denominators and explains why association tests are not appropriate for an outcome with no variation.

Study	9 methods × 4 scenarios × 10 paired repetitions
Navigation data	360 valid runs; 360 successful; 0 excluded; 0 failed
Source	Corrected manuscript supplied for this update (13 September 2026)

Agri-Social Navigation in Vineyard Corridors

1. Outcome rule

A run is successful when it completes the A -> B -> A mission within the 180 s limit. Accounting separates planned, executed, valid, excluded, successful, and failed runs so that file integrity is not confused with operational outcome.

2. Method-level accounting

Method	Planned	Executed	Valid	Excluded	Success	Failed
M0	40	40	40	0	40	0
M1	40	40	40	0	40	0
M2	40	40	40	0	40	0
M3	40	40	40	0	40	0
M4	40	40	40	0	40	0
B1	40	40	40	0	40	0
B2	40	40	40	0	40	0
B3	40	40	40	0	40	0
B4	40	40	40	0	40	0
Total	360	360	360	0	360	0

Each method contains four scenarios with 10 planned, executed, valid, and successful runs per scenario. The design has 36 Method × Scenario cells, none with an exclusion or failure.

3. Statistical consequence

$$\eta_{\text{success}} = N_{\text{success}} / N_{\text{valid}} = 360 / 360 = 1$$

Because every method has 100% success, there is no between-group variation. A chi-square or Fisher test provides no discrimination and degenerates because no failures are observed. The appropriate action is to report the constant result and remove success from TOPSIS.

4. What can and cannot be concluded

Supported:

- Automation completed the planned matrix without exclusions.
- Continuous endpoints are compared using the same $n = 40$ per method.
- Complete mission completion removes the bias of comparing descriptives with different denominators.

Not supported:

- It does not prove that the true failure probability is zero.
- It does not distinguish safety, comfort, time, path length, or smoothness.
- It is not validation on a physical robot or with real workers.

5. Archive control

The matrix should be audited using the unique Method-Scenario-RunID key and cross-checked against raw CSV, metrics CSV, and MAT files. The count of 360 belongs to the navigation campaign; the 54 computational runs form a separate benchmark.